

■ COORDINATE GEOMETRY

Plotting Points

Planrs Academy • Session 1 • Beginner

■ Across First, THEN Up! ■

Welcome, young navigator! In this lesson you will learn to read and plot points on a **coordinate grid** — the same kind of map that sailors, pilots, and treasure hunters use to find exact locations. Every point on the grid has its own address, written as **(x, y)**. By the end of this session you will be able to label the axes, plot points, and even trace hidden shapes!



THE 5 KEY TERMS YOU WILL MASTER

Term	Pronunciation	Meaning	Axis Colour
Grid	GRID	Criss-cross lines making squares on a map	—
Origin	OR-ih-jin	Starting point (0, 0) where axes meet	★ Gold dot
x-axis	EX-ax-is	Horizontal line (goes LEFT → RIGHT)	● Blue
y-axis	WHY-ax-is	Vertical line (goes UP ↑)	● Red
Coordinate Pair	koh-OR-din-it	Two numbers (x, y) giving an exact location	—

Learning Objectives

LO1 — **IDENTIFY** the x-axis and y-axis on a grid.

LO2 — **READ** a coordinate pair (x, y) using "across first, then up."

LO3 — **PLOT** at least 5 points using whole numbers 0–5.

LO4 — **DESCRIBE** a point using the words: x-coordinate, y-coordinate, origin.

LO5 — **APPLY** coordinates to complete a connect-the-dots activity.

1 WHAT IS A COORDINATE GRID?

Concept 1: The Grid — Our Special Map

Definition: A grid is a set of lines crossing each other to make squares, like the squares in a ruled notebook.

Example: Think of your school attendance register. Rows go left to right, columns go up and down. Every box has an exact address — like Row 3, Column 5. Our grid works the same way!

Memory Trick: Picture the squares on your Classmate notebook. Each little box is one "step" on our grid.

2 THE ORIGIN — OUR STARTING POINT (0, 0)

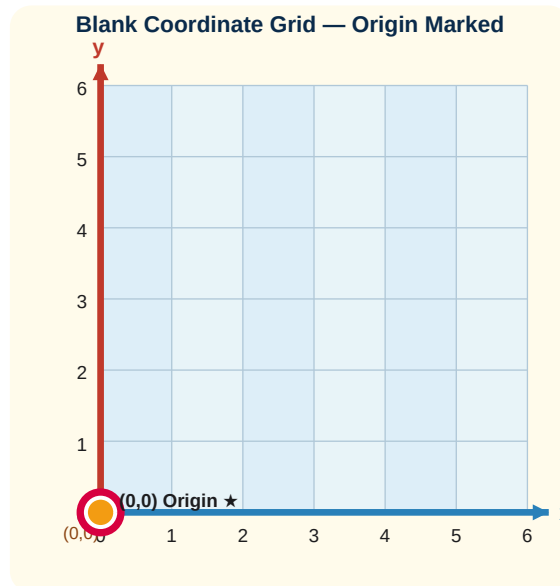
Concept 2: Origin (0, 0)

Definition: The origin is the point (0, 0) — the corner where the x-axis and y-axis meet. All counting starts here.

Example: Imagine you are standing at your school gate. That is (0,0). Walk 3 steps toward the playground ($x = 3$), then 2 steps toward the water tap ($y = 2$). You are at (3, 2)!

Memory Trick: The origin is like India Gate in Delhi — all roads are measured FROM that zero point.

Here is a blank grid — can you spot the origin (0, 0)?



■ The origin is ALWAYS at position (0, 0).

3 THE X-AXIS — THE ROAD GOING ACROSS

Concept 3: x-axis (Horizontal Line)

Definition: The x-axis is the HORIZONTAL line. It goes LEFT to RIGHT across the grid. The x-coordinate tells us HOW FAR ACROSS a point is.

Example: Imagine the National Highway going from Mumbai to Hyderabad — it stretches across the map. That is our x-axis! The x-axis is always drawn in BLUE.

Memory Trick: X is like a road you drive ON — flat, going side to side. Think: eXpress highway going across the country!

4 THE Y-AXIS — THE ROAD GOING UP

Concept 4: y-axis (Vertical Line)

Definition: The y-axis is the VERTICAL line. It goes UP and DOWN. The y-coordinate tells us HOW HIGH UP a point is.

Example: Think of a multi-storey building. The lift moves UP and DOWN. Floor 0 = ground floor, Floor 4 = fourth floor. The y-axis is always drawn in RED.

Memory Trick: Y goes UP like a rocket — Y is WHY the rocket goes so HIGH. Think: Y = Yonder (far up in the sky)!

■ x-axis Quick Facts

- Direction: LEFT → RIGHT (horizontal)
- Colour in this lesson: BLUE
- Counts: how far ACROSS
- Memory: eXpress highway!

■ y-axis Quick Facts

- Direction: UP ↑ (vertical)
- Colour in this lesson: RED
- Counts: how far UP
- Memory: Y = Yonder (sky high)!

5 COORDINATE PAIRS — THE ADDRESS SYSTEM (X, Y)

Concept 5: Coordinate Pair (x, y)

Definition: A coordinate pair is TWO numbers written like (3, 2). The FIRST number is the x-coordinate (how far across). The SECOND number is the y-coordinate (how far up).

Example: Your home address has: Street Number first, then Floor/Flat Number. "Plot 5, Floor 2" = (5, 2) on our grid. We always say the street (ACROSS) before the floor (UP).

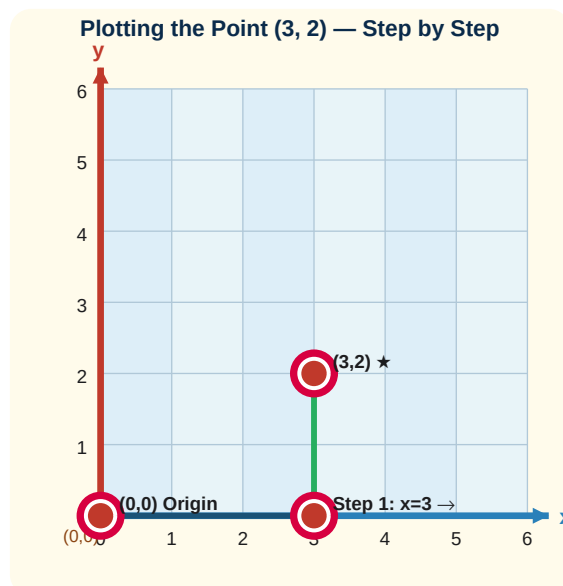
Memory Trick: Go along the corridor, THEN go up the stairs! OR: Walk RIGHT first, then CLIMB UP.

Across first (x), THEN Up (y)! Always!



STEP-BY-STEP: HOW TO PLOT POINT (3, 2)

Step 1	■ Start at the origin (0, 0) — the corner where both axes meet.
Step 2	➡■ Read the FIRST number: $x = 3$. Count 3 steps to the RIGHT along the x-axis.
Step 3	■■ Read the SECOND number: $y = 2$. Count 2 steps UP from where you stopped.
Step 4	⇒■ Mark a dot at that exact spot. Write the label (3, 2) next to it.
Step 5	■ Check: Did you go ACROSS first, then UP? That is the golden rule!



The dashed path shows: go RIGHT 3 steps, then UP 2 steps. The red dot at (3, 2) is the treasure!

**COMMON MISTAKE — DON'T FLIP X AND Y!****Wrong Way**

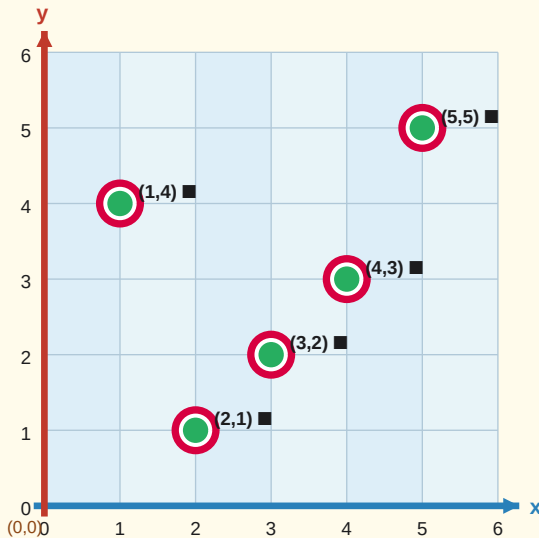
For point (4, 1):
Some students count **UP 4**, then **RIGHT 1**.
This puts the dot in the **WRONG place!**

Rule broken: y used before x.

Correct Way

For point (4, 1):
Count **RIGHT 4** first ($x = 4$),
then **UP 1** ($y = 1$).
CORRECT!

Rule followed: x first, then y.

**GUIDED PRACTICE — PLOT THESE 5 POINTS****Practice Grid — 5 Points Plotted****Points to Plot:**

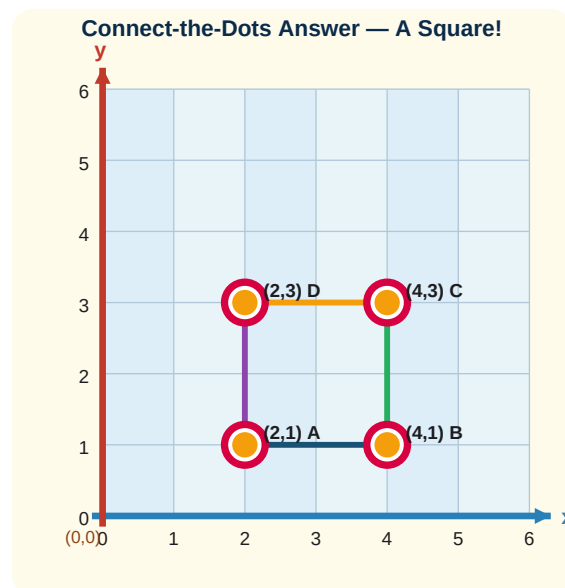
1. (1, 4) — Anchor ■
Go RIGHT 1, UP 4
2. (3, 2) — Diamond ■
Go RIGHT 3, UP 2
3. (5, 5) — Parrot ■
Go RIGHT 5, UP 5
4. (2, 1) — Key ■
Go RIGHT 2, UP 1
5. (4, 3) — Flag ■
Go RIGHT 4, UP 3

ACROSS first, THEN UP!

**FUN CHALLENGE — CONNECT THE DOTS!**

First **plot** each point, then **connect** them in order to reveal a hidden shape. Follow the sequence carefully — it matters!

Step 1	■ Plot (2, 1) — START HERE
Step 2	■ Plot (4, 1)
Step 3	■ Plot (4, 3)
Step 4	■ Plot (2, 3)
Step 5	⇒■ Connect (2,1) → (4,1) — draw a line
Step 6	⇒■ Connect (4,1) → (4,3) — draw a line
Step 7	⇒■ Connect (4,3) → (2,3) — draw a line
Step 8	⇒■ Connect (2,3) → (2,1) — draw a line back to the start
Step 9	■ What shape did you make? (Hint: it has 4 equal sides!)



The four points (2,1), (4,1), (4,3), (2,3) form a perfect square!

**STUDENT WORKSHEET — TRY IT YOURSELF!**

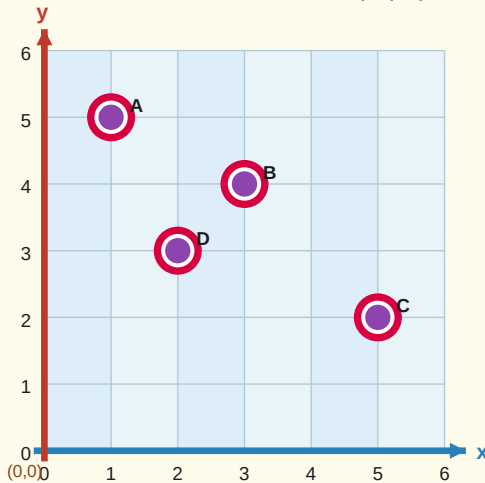
Name: _____ Date: _____ Score: _____ / 10

Part A — Fill in the Blanks (1 mark each)

1. The horizontal line is called the _____.
2. The vertical line is called the _____.
3. The two axes meet at a point called the _____ at (0, 0).
4. In the pair (4, 3), the number 4 is the _____-coordinate.
5. In the pair (4, 3), the number 3 is the _____-coordinate.
6. We always move _____ first, then _____ up.

Part B — Name the Coordinates (1 mark each)

Look at each plotted point below and write its coordinate pair:

Name the Coordinates of Points A, B, C, D

Point A = (__ , __)

Point B = (__ , __)

Point C = (__ , __)

Point D = (__ , __)

**ANSWER KEY**

Question	Answer
1. Horizontal line	x-axis
2. Vertical line	y-axis
3. Where axes meet	origin
4. First number in (4,3)	x
5. Second number in (4,3)	y
6. Direction rule	ACROSS first, then UP

Point	Answer
A	(1, 5)
B	(3, 4)
C	(5, 2)
D	(2, 3)

**TEACHER & PARENT TIPS****■ Try at Home — Chessboard**

Place a chess/carrom board flat.

Call one corner (0,0). Count squares across = x, up = y.

Call out coordinates and ask the child to point to that square!

■ Try at Home — Notebook Grid

Open a square-ruled notebook.

Mark the bottom-left corner as (0,0).

Ask the child to plot (2,3) and (4,1) using a pencil dot.

■ Try at Home — Floor Tiles

Stand at a room corner — that is (0,0).

Walk 3 tiles right, then 2 tiles forward.

"You are now at (3,2)!" — makes coordinates feel real.

■ Most Common Error to Watch For

Students often read y before x — they go UP first, then across.

Remind them: "Go along the corridor FIRST, then climb the stairs."

Use physical movement: walk sideways (x), then jump up (y).

■ Across first (x), THEN Up (y) — ALWAYS! ■